

Rajanie Prabha

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SKILLS

AI for Sustainability, Machine Learning, Computer Vision, Remote Sensing ML, Vision-Language Models, Data Analytics, Power Grid Optimization, Python, PyTorch, TensorFlow

EDUCATION

Stanford University, PhD, AI for Sustainability Oct 2021—May 2026
Technical University of Munich, Master of Science, Computer Science (ML track) Oct 2017—Nov 2019
National Institute of Technology Kurukshetra, Bachelor of Technology, Computer Science Aug 2012—July 2016

SELECTED PUBLICATIONS

1. **R. Prabha**, L. Min, R. Rajagopal. *Peak-summer N-1 analysis reveals widespread headroom and localized bottlenecks in the Western US transmission grid*, Under Submission
2. **R. Prabha**, C. Zanocco, Z. Wang, J. Flora, R. Rajagopal. *Where Solar Surges Next: Distribution-Planning Forecasts from Zero-Inflated Neural Models*, Accepted at ACM E-Energy, 2026
3. **R. Prabha**, Z. Wang, C. Zanocco, J. Flora, R. Rajagopal. *Nationwide Insights on Solar Deployment Trends and Spatial Equity Using Vision Transformer Models*, Applied Energy, 2026.
4. **R. Prabha**, Z. Wang, C. Zanocco, J. Flora, R. Rajagopal. *DeepSolar-3M: An AI-Enabled Solar PV Database Documenting 3 Million Systems Across the US*, ICLR Climate Change AI Workshop, 2025. [Best Paper Award; 1 in 81]
5. M. Wussow, C. Zanocco, Z. Wang, **R. Prabha**, J. Flora, D. Neumann, A. Majumdar, R. Rajagopal. *Exploring the Potential of Non-Residential Solar to Tackle Energy Injustice*, Nature Energy, 2023.
6. Z. Wang, **R. Prabha**, T. Huang, J. Wu, R. Rajagopal. *SkyScript: A Large and Semantically Diverse Vision-Language Dataset for Remote Sensing*, AAAI 2024.
7. P. Buitrago, E. Toropov, **R. Prabha**, J. Uran, R. Adal. *MiikeMineStamps: A Long-Tailed Dataset of Japanese Stamps via Active Learning*, ICDAR, Conference Paper, September 2021.
8. **R. Prabha**, M. Tom, M. Rothermel, E. Baltsavias, L. Leal-Taixe, K. Schindler. *Lake Ice Monitoring with Webcams and Crowd-Sourced Images*, ISPRS Conference Paper, August 2020.

RESEARCH EXPERIENCE

Stanford University, Graduate Research Assistant in Prof. Ram Rajagopal's Lab Oct 2021—Present

- Led a large-scale analysis of transmission grid utilization for the Western Electricity Coordinating Council (WECC), assessing over 35,000 transmission facilities under N-1 contingency optimization conditions; Proficiency in TARA by PowerGEM. [Arcgis storymap]
- Created a large image-text dataset for remote sensing with 2.6 million image-text pairs covering 29K distinct semantic tags leveraging CLIP VLM. [Github]
- Conceptualized and implemented a Vision transformer based data acquisition pipeline (with LoRA for computational efficiency) that spatiotemporally tracks and monitors solar PVs across the US at the building level with over 90% accuracy. [Github]

Pittsburgh Supercomputing Center, Machine Learning Research Scientist Feb 2020—Sept 2021

- Benchmarked and performed a comparative evaluation of various AI hardware: NVIDIA Pascal, Voltas, DGX-2 for Maskrcnn and Single Stage Detection Models.
- Supervised a project on active learning with the University of Pittsburgh as an Extended Collaborative Support Service ML expert under Extreme Science and Engineering Discovery Environment (XSEDE).
- Reviewed proposals, created user guide, and provided Machine Learning assistance to 10 projects across the US involved with the Early User Program of Neocortex, the AI supercomputer built on Cerebras chips.
- Developed and ran a workshop on Nvidia profilers, singularity containers, Scaling best practices, and related research findings.

ETH Zurich, Student Researcher, advised by Prof. Konrad Schindler Apr 2019—Nov 2019

- Designed and implemented a semantic segmentation model Deep-U-Lab to detect ice on lakes in Switzerland.

PROFESSIONAL EXPERIENCE

[Google] **X**, Moonshot factory, Mountain View, AI PhD Resident Jun 2024—Aug 2024

- Worked on asset management for **Tapestry**, X's moonshot for the electric grid. Trained a computer vision model for detecting the electrical properties of pole-mounted transformers with 90% accuracy.

Philips Innovation Center, Bangalore, Software Engineer Jun 2016—Jul 2017

- Maintained various features like remote accessibility, log collection, monitoring rules, data sources, and regular expressions on the Serviceability Platform for various medical devices and resolved around 50 customer requests. Awarded Philips Recognition Certificate for handling Computed tomography and Magnetic resonance imaging modalities.

Luminovo.AI, Munich, Research Intern

Apr 2018—Sept 2018

- Implemented Tacotron 2 paper (Text-to-Speech Synthesis) on the English LJ speech dataset and German Angela Merkel dataset (M-AILABS).

AWARDS

- Best Paper Award for *DeepSolar-3M* at the ICLR 2025 Workshop: Tackling Climate Change with Machine Learning
- IEEE Conference on Secure and Trustworthy ML (SaTML) travel award, 2022
- International Conference on Learning Representations (ICLR) conference grant, 2021, 2022
- WiML NeurIPS conference grant, 2020
- Swiss European Mobility Programme Scholarship, 2019
- Inspire Scholarship, DST, Government of India, 2012 [Top 1 percentile]

TALKS, OUTREACH & SERVICE

- **Reviewer:** Big Earth Data Journal, ML Reproducibility Challenge, Advances in Approximate Bayesian Inference (AABI) @ ICLR, ICBINB @ NeurIPS, ML for Physical Sciences @ NeurIPS, Grace Hopper Celebration (GHC)
- **Invited Talks:**
 - **SkyScript** - Earth-ML Lunch, Stanford (May 2025)
 - **DeepSolar-3M** – Sustainability Data Science Conference, Stanford (Apr 2025)
 - **SUSTAIN 210M** - Guest Lecture, Stanford (May 2024)
 - **Generative Editing for Adversarial Attacks** – Poster, EEML (2023)
 - **CNN for Dummies** – AnitaB.org (Apr 2021)
 - **Explainability in AI** – AnitaB.org (Jul 2021)
- **Teaching Assistant**, ENERGY 276: Electric System Planning with Emerging Generation Technologies and Large Load, Stanford (Fall 2025)
- **Graduate Mentor**, AI4ALL, Stanford (2025)
- **Leadership & Service:**
 - Women Techmakers Ambassador, Google (2022–2025)
 - Women Techmakers Member, Google (2021–2022)
 - Member, AI Committee, AnitaB.org (2019–2024)